MADAN MUSUVATHI

Founder & CEO, Truidic | ACM Fellow | Former Partner Research Manager, Microsoft Research | Stanford Ph.D.
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PROFILE

Founder, researcher, engineering leader, and ACM Fellow with 25+ years of experience turning foundational formal verification research into industrial-scale systems. Pioneered scalable model checking and concurrency verification; built tools deployed across Microsoft and the software industry; and developed formal-methods-driven optimizations for production AI infrastructure. Recruited and led Microsoft Research's RiSE group, which advanced influential technologies including Lean, Z3, F*, and TLA+. Recipient of the 2023 CAV Award.

WHY EVOLVE @ 8090

I strongly believe AI is a force for good for humanity, but only if we make it safe and secure. This conviction led me to leave Microsoft after 22 years and found Truidic, where we are building industrial-scale formal verification technology for AI systems. At EVOLVE, I want to integrate Truidic's verification capabilities with 8090's Software Factory so AI coding agents can generate code, identify vulnerabilities, repair defects, and verify specified security properties with mathematical guarantees. My goal is to demonstrate a path toward enterprise AI development where security is verified rather than merely assumed.

EXPERIENCE

TRUIDIC | Founder & CEO | Sep 2026–Present

- **Founded Truidic** to make AI-generated software secure through industrial-scale formal verification.
- **Developing verification technology** combining program analysis, symbolic reasoning, and AI-assisted automation to discover vulnerabilities and establish mathematically backed security guarantees.
- **Leading** research, product engineering, early customer engagement, and partnerships with AI software-development platforms.

MICROSOFT RESEARCH | Partner Research Manager; Researcher & Research Leader | 2004–2026

- **Research leadership.** Recruited, developed, and led RiSE, a world-leading research group in formal methods, programming languages, and software engineering; set long-term strategy, developed talent and supported foundational projects including Lean, Z3, F*, and TLA+.
- **Industrial-scale verification.** Pioneered scalable concurrency testing and verification algorithms for real-world systems. Shipped tools for user-mode and kernel-mode software, including integrations with Application Verifier and Driver Verifier. Work helped uncover thousands of concurrency defects and influenced subsequent approaches to race detection, including Linux KCSAN. Received the 2023 CAV Award.
- **AI infrastructure optimization.** Developed formal-methods-based techniques and systems including MSCCL and MSCCL++ to optimize GPU communication kernels. Optimized communication algorithms were used in production AI workloads, including Azure OpenAI inference; MSCCL is incorporated into AMD RCCL.
- **AI-assisted software engineering.** Advanced research on natural-language specifications, intent formalization, automated bug finding, and performance optimization.

INTERPROPHET | Founding Engineer | 1997–1998

- **Built** a hardware-accelerated TCP/IP stack for low-latency Internet applications at an early networking startup.

EDUCATION

STANFORD UNIVERSITY | Ph.D. and M.S., Computer Science | Ph.D. 2004

Pioneered scalable software model checking, developing one of the first model checkers capable of booting Linux and verifying components of a real operating system. Developed techniques for verifying complex C/C++ software.

INDIAN INSTITUTE OF TECHNOLOGY MADRAS | B.Tech., Computer Science

SELECTED HONORS & SERVICE

ACM Fellow • 2023 CAV Award • Best Paper Awards: OSDI 2004, SOSP 2019, PPoPP 2021 • Program Co-Chair: ASPLOS 2024, PPoPP 2026 • Steering Committees: ASPLOS, PPoPP • 150+ peer-reviewed publications; 10k+ citations; 48 h-index • 20+ patents